⚠ EARLY-VERSION CONCEPT ILLUSTRATION · Ø3 m BORE DESIGN · NOT FOR CONSTRUCTION · ALL DIMENSIONS PRELIMINARY ⚠

*SELENE design map, companion to the BRONTE Chimborazo sheet (KER-BRO-TRK-001). One machine, one ridge: **EOS-3** (Ø3.0 m bore, Ø2.6 m pod, 7 t bulk per shot). Bore fixed across all five phases; only the track grows. Phase 1b baseline: 2.9 km at 100 g, escape speed 2.38 km/s, 8.7 MW south-pole solar. Total build estimate: $2.0B–$2.5B (mid-case, see Panel F). Mission one: cargo and satellite dispensers launched from the Moon and returned to Earth or delivered to cislunar orbit. The site map uses real NASA LRO/LOLA laser topography (20 m/px polar DEM); track alignment is conceptual. All physics re-derived June 2026.*

ANNEX A

SELENE, South Pole Design Map

KER-SEL-TRK-002 · REV E · Ø3 m BORE · EARLY-VERSION TEST MOCK-UP · NOT FOR CONSTRUCTION

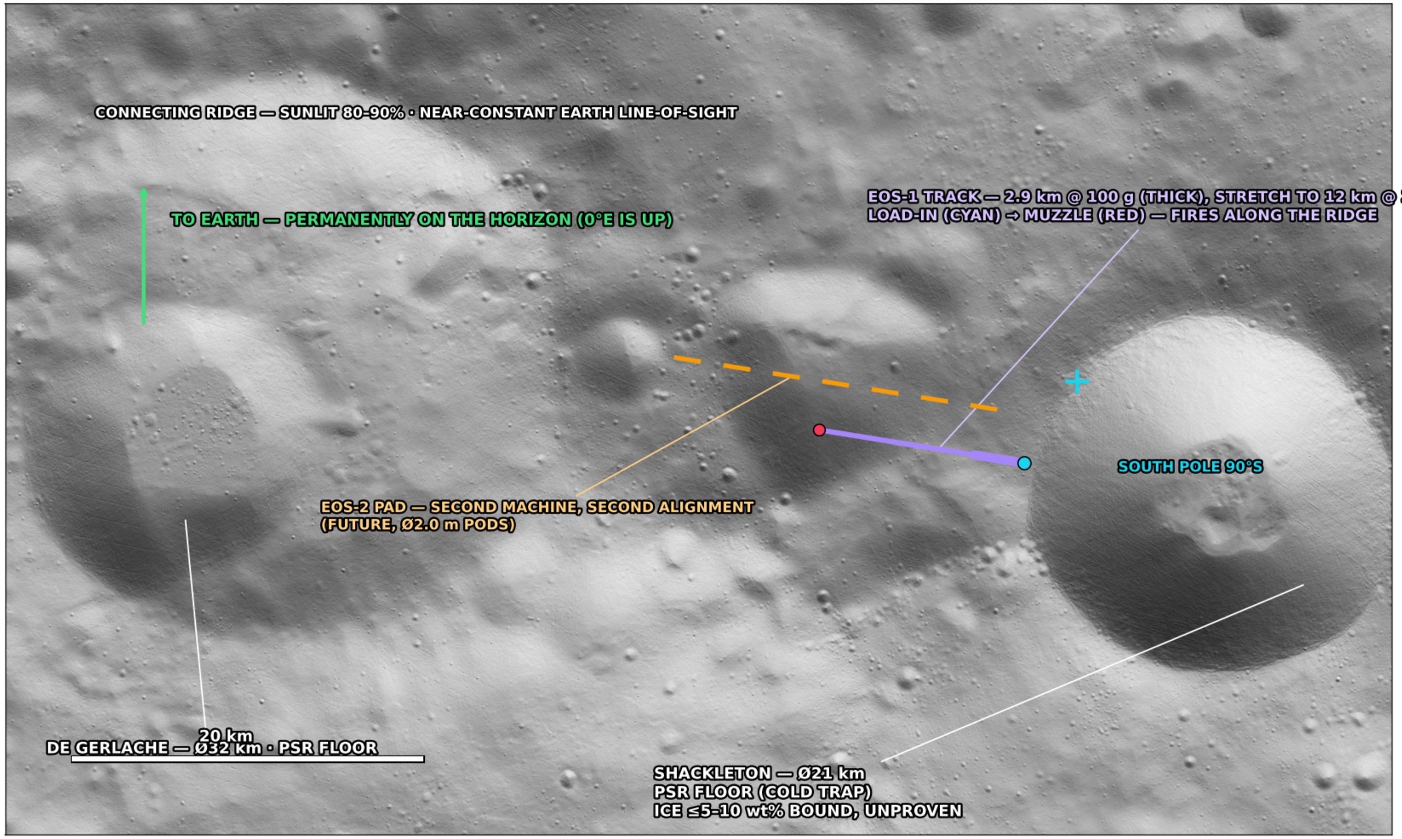
WHAT IS INSIDE

- Panel A, Site: real LRO/LOLA laser topography, Shackleton / Connecting Ridge ~89.7°S
- Panel A2, Where on the Moon · what does not exist here · EOS-3 machine specs (Ø3 m bore)
- Panel B, EOS-3 open-track section: no tube, the Moon is the vacuum vessel (Ø3.0 m bore)
- Panel C, Side profile: track on an open strut trestle, steeper exit angle (true 1:1)
- Panel D, Engineering charts: track length vs g, Δv destinations, ridge illumination, energy
- Panel E, Return & delivery: the Earth–Moon system, Lagrange points, and how orbits actually work
- Panel F, Build cost summary (revised Ø3 m estimate, $2.0B–$2.5B mid-case)

Each large panel is on its own page, rotated to read at full size, turn the page (or your screen) a quarter-turn clockwise.

PANEL A, SITE: REAL LRO/LOLA TOPOGRAPHY, SHACKLETON / CONNECTING RIDGE, ~89.7°S

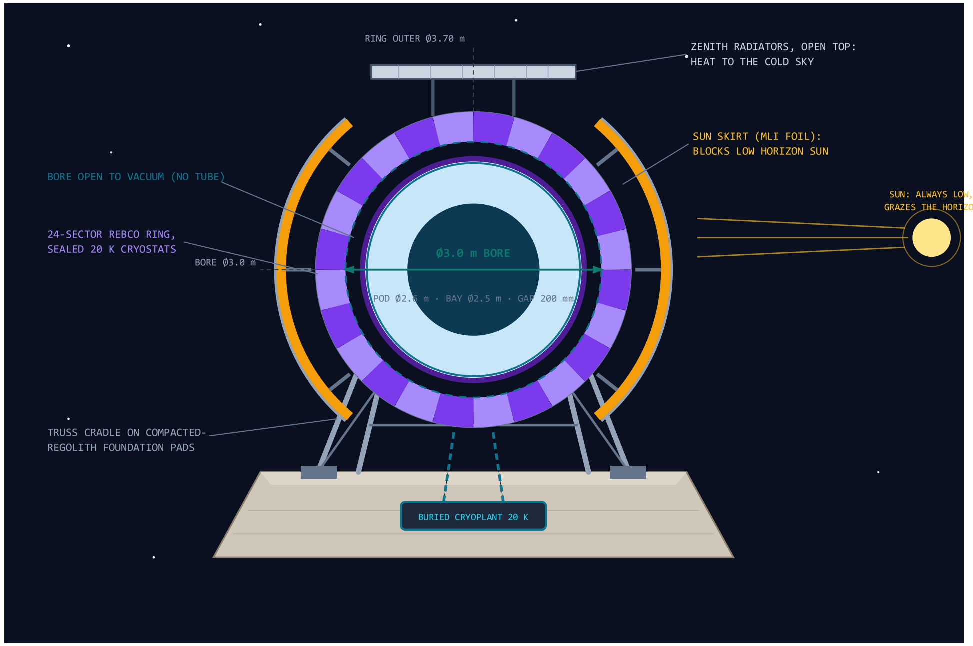
LUNAR SOUTH POLE — REAL LRO/LOLA TOPOGRAPHY (20 m/px, NASA GSFC) · HILLSHADE RENDER · EARLY CONCEPT, NOT FOR CONSTRUCTION



PANEL B, EOS-3 OPEN-TRACK SECTION, NO TUBE: THE MOON IS THE VACUUM VESSEL — Ø3.0 m BORE

EOS-3 SECTION · Ø3.0 m BORE · POD Ø2.6 m · GAP 200 mm · PAYLOAD BAY Ø2.5 m

SCALE: ~80 px/m · 24-SECTOR REBCO RING @ 20 K IN SEALED CRYOSTATS · POLAR SUN SKIRTS BLOCK LOW HORIZON SUN · ZENITH RADIATORS · RING OUTER Ø3.70 m



FOUNDATION SET IN REGOLITH ~250 K BELOW 2 m: A NATURAL COLD SINK
RING OUTER Ø3.70 m (BORE Ø3.0 m + 0.35 m WALL EACH SIDE)

POD "EOS-3" · Ø2.6 m × 5 m · 3 MISSION CONFIGS

CONFIG A — BULK CARGO:
PELLETS (REGOLITH/ICE/METAL): 7,069 kg = 14.1 m³ BAY @ 500 kg/m³

CONFIG B — PACKAGED CARGO:
3 SWAP RACKS · PRESSURE VESSELS · MANUFACTURED GOODS
~3,000–5,000 kg USEFUL LOAD · RCS FOR ORBIT TRIM

CONFIG C — SATELLITE DISPENSER:
~15–20 × 100 kg SMALLSATS IN Ø2.5 m × 2.9 m BAY · DISPENSE AFTER RELEASE
ONBOARD Δv: ~200 m/s LUNAR ORBIT/L1/L2 · ~800 m/s GEO · LEO NEEDS KICK STAGE

ALL 3 CONFIGS: SAME RING, SAME TRACK, SAME DEPARTURE SPEED
HARDENED TO 100 g · ARTILLERY ELECTRONICS: 15,000–20,000 g QUAL
PICA HEAT SHIELD: DUST DOME AT LAUNCH, ENTRY SHIELD AT EARTH

THE HONEST HARD PARTS (LUNAR EDITION, Ø3 m)

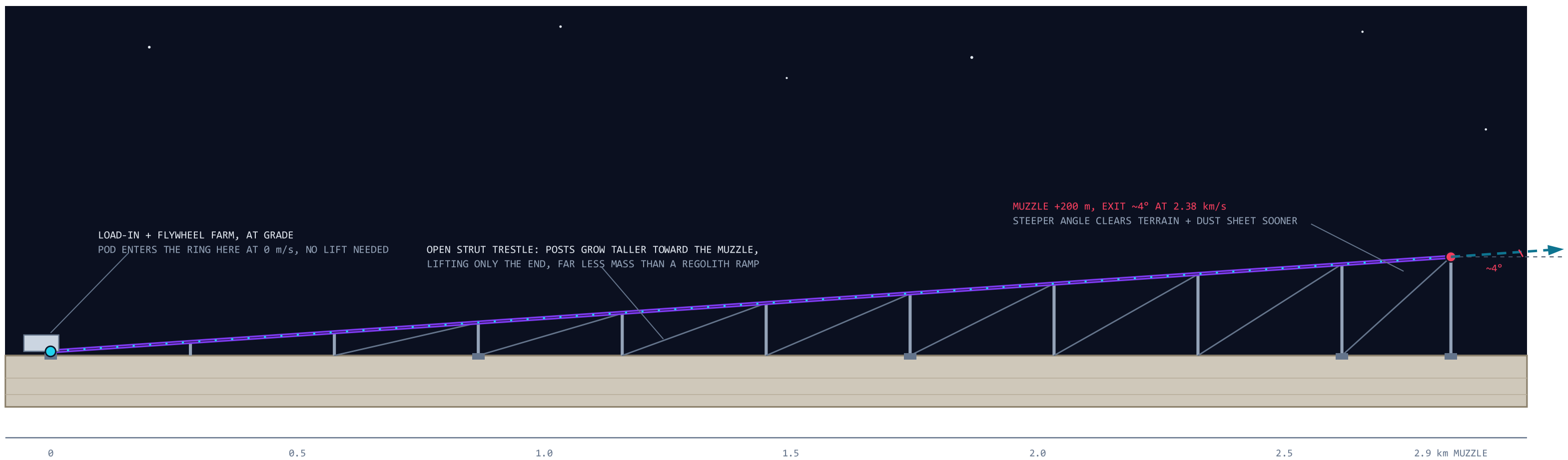
1. DUST. EJECTA LOFTED AT POD RELEASE; ELECTROSTATIC ON RIDGE. DOME, PLINTHS, EDS-COATED ARRAYS — ALL EXIST BECAUSE OF DUST.
2. THERMAL. 280° SHIELD + SEALED CRYOSTATS + BURIED CRYO LOOP. 390 K SUN TO CRYO-COLD SHADOW ON ONE TRACK, UNAVOIDABLY.
3. THE PULSE. 0.79 MWh IN 2.4 s = 2.37 GW. FLYWHEEL FARM CHARGED SLOWLY BY 8.7 MW SOLAR, DUMPED FAST. MASS LANDS FIRST.
4. CATCHING. MISSION ONE: EARTH ATMOSPHERE IS THE MITT (SHIELD + CHUTES). CISLUNAR RENDEZVOUS IS THE NEXT PROBLEM.
5. ICE UNCERTAINTY. SHACKLETON ≤5–10 wt%, UNCONFIRMED. REGOLITH + METAL CLOSE THE EARLY BUSINESS WITHOUT ICE.

MISSING FROM THIS LIST IS THE POINT: NO TUBE, NO WINDOW, NO BOOM, NO HEATING, NO WEATHER, NO RESERVE LAWYERS, NO FUEL.

PANEL C, SIDE PROFILE: TRACK ON AN OPEN STRUT TRESTLE, STEEPER EXIT (TRUE SCALE 1:1)

EOS-3 PoC TRACK: 2.9 km · LOAD-IN AT GRADE → MUZZLE +200 m ON AN OPEN STRUT TRESTLE · EXIT ~4°

TRUE SCALE 1:1 (0.47 px/m) · THE EXIT ANGLE IS SET BY STRUT HEIGHT AT THE MUZZLE END, NOT BY A GROWING REGOLITH BERM



TRUE SCALE 1:1 · STRAIGHT TRACK (NO CURVE, NO ADDED g) · THE MUZZLE +200 m IS LIFTED BY OPEN STRUTS, SO THE EXIT ANGLE IS TUNABLE WITHOUT MOVING REGOLITH

PANEL A2, WHERE ON THE MOON · WHAT DOES NOT EXIST · EOS-3 MACHINE SPECS

WHERE ON THE MOON?

HERE, SOUTH POLE (89.7°S)

PHOTO: REAL NEARSIDE · MAP: REAL LRO/LOLA LASER DATA
SHACKLETON RIM + CONNECTING RIDGE: 80–92% ANNUAL SUN

WHAT DOES NOT EXIST ON THIS SHEET

NO VACUUM TUBE — THE TRACK IS ALREADY IN HARD VACUUM

NO PLASMA WINDOW — NO ATMOSPHERE TO KEEP OUT

NO AERO-SHELL — NO NEEDLE NOSE — NO AIR TO CUT THROUGH

NO ISOLATION GATES — NO BLAST SHUTTERS — NO TUBE BREACH

NO SONIC BOOM — NO LAUNCH HEATING — NO DRAG, EVER

EVERY HARD SUBSYSTEM ON THE BRONTE SHEET IS DELETED BY ONE SENTENCE: THE MOON HAS NO ATMOSPHERE.
WHAT REMAINS IS THE RING, THE POWER, AND THE COLD.

WHAT REPLACES THEM: DUST + 250–390 K THERMAL SWINGS + GW-CLASS PULSE STORAGE + CATCHING THE CARGO AT THE OTHER END.

SITE SERVICES: RIDGE SOLAR 80–92% UPTIME · PSR COLD SINK NEXT DOOR · REGOLITH ~250 K AT 2 m DEPTH, NATURAL CRYO BUFFER

MAP DATA: NASA LRO/LOLA POLAR DEM, 20 m/px (LDEM_875S_20M), POLAR STEREO, RENDERED AS HILLSHADE. A REAL SURVEY.

CHOSEN BORE: Ø3 m. SAME RING ACROSS ALL 5 PHASES.
TRACK GROWS; BORE DOES NOT CHANGE.

EOS-3 · Ø3 m BORE · THE CHOSEN MACHINE

PHASE 1b PoC: 2.9 km · POD Ø2.6 m · BORE Ø3.0 m · GAP 200 mm
7,069 kg BULK PER SHOT (14.1 m^3 BAY @ 500 kg/m^3) · 3,000–5,000 kg CARGO
ENERGY 0.79 MWh/SHOT · PEAK ~2.37 GW · RECHARGE ~5 min @ 8.7 MW
~100 SHOTS/HR BULK OR ~10 LARGE CARGO SHOTS/HR

THROUGHPUT: ~72,000 t/yr AT 8.7 MW (POWER-LIMITED, NOT BORE-LIMITED)
THROUGHPUT RULE: 8,378 t/MW/yr AT ESCAPE SPEED
SAME TONNAGE ALL BORE SIZES: BORE BUYS SHOT MASS, NOT ANNUAL MASS

REBCO: 2,809 km TAPE · 6.7 t · $281M @$100/m · $13M DELIVERY
24-SECTOR RING @ 20 K, 0.6 m PITCH · 4,833 RINGS OVER 2.9 km
ALL 5 PHASES USE IDENTICAL RING: TRACK EXTENDS, RING STAYS

SCALING LADDER: 2.9 km → 12 km → 30 km → 50 km → 96 km
ESCAPE (2.38) · CARGO-+ (2.66) · HIGH-SPD (4.20) · LLO/3g · ESC/3g

FIRST BILL IS A ROCKET BILL: 6.7 t REBCO + ELECTRONICS + SOLAR PLANT
CO-MANIFESTED ON 2–3 STARSHIP LANDER FLIGHTS · ~$254M DELIVERY

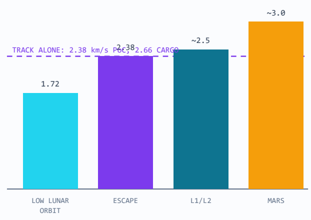
RETURN MISSIONS: CARGO, SAMPLES, SATELLITE DISPENSERS (SEE PANEL E)
POD'S PICA HEAT SHIELD DOUBLES AS DUST DOME AT LAUNCH AND ENTRY SHIELD FOR DIRECT ATMOSPHERIC RECOVERY AT EARTH.
BUILD ESTIMATE: $2.0B–$2.5B MID-CASE (SEE PANEL F)

PANEL D, ENGINEERING CHARTS: TRACK LENGTH · DESTINATIONS · ILLUMINATION · ENERGY

TRACK LENGTH FOR ESCAPE vs ACCELERATION

$L = v^2/2a$ AT v = 2.38 km/s. ACCELERATION CHOICE IS THE ENTIRE COST MODEL.

WHERE THE TRACK GETS YOU (Δv FROM SURFACE)



SMALL ONBOARD Δv (200–400 m/s) TOPS UP TO ANY CISLUNAR ORBIT OR EARTH RETURN.

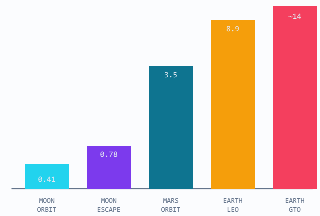
RIDGE ILLUMINATION OVER A YEAR (SCHEMATIC)

GOLD = SUNLIT (80–92% ANNUAL AT RIDGE SITES)

DARK SPELLS: HOURS TO 3–5 EARTH-DAYS MAX. BRIDGED BY THE SAME FLYWHEEL FARM THAT FIRES THE TRACK. LAUNCHES PAUSE, NO FREEZE.

LUNAR AXIAL TILT ONLY 1.5°. POLAR RIDGES ARE THE MOST THERMALLY STABLE GROUND ON THE MOON. (GLÄSER ET AL. 2014, LRO/LOLA)

ENERGY/kg: WHY THE MOON WINS

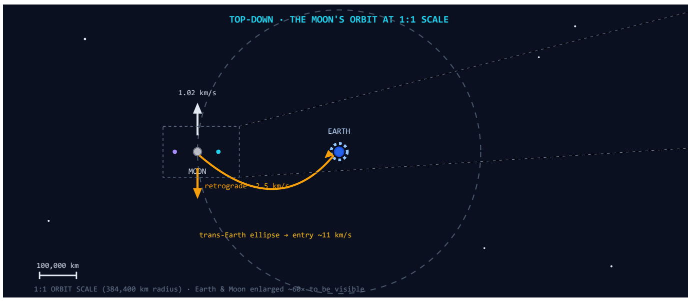


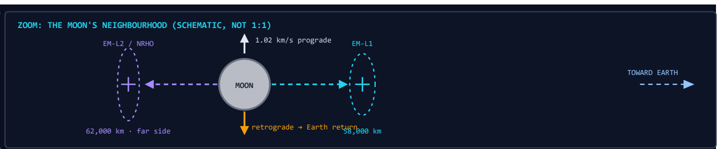
kWh/kg. MOON ESCAPE = 1/11th OF EARTH-TO-LEO. EVERY DEPOT, STATION, OR SHIPYARD GETS CHEAPER MASS FROM THE MOON THAN FROM EARTH.

PANEL E, RETURN & DELIVERY: THE EARTH–MOON SYSTEM AND HOW ORBITS ACTUALLY WORK

THE EARTH–MOON SYSTEM: LAUNCH DIRECTION, NOT JUST SPEED, SETS THE DESTINATION

Left: the Moon's orbit at true 1:1 scale (bodies enlarged ~60× to be visible) · Right: a zoom of the Moon's neighbourhood — EM-L1 ~58,000 km toward Earth, EM-L2 ~62,000 km beyond the far side





THE PRINCIPLE: DIRECTION, NOT JUST SPEED

Escape speed (2.38 km/s) alone just leaves the payload in the Moon's own ~1.02 km/s orbit around Earth — it shadows the Moon, it does not fall to Earth on its own.

Direction sets the destination: launch RETROGRADE to drop the transfer-ellipse perigee toward Earth; launch PROGRADE / faster to climb to the L-points, GEO, or escape.
EM-L1 sits toward Earth; EM-L2 sits beyond the far side.

① RETURN TO EARTH SURFACE

samples · He-3 · ZBLAN fiber · pharma crystals

- Launch retrograde to the Moon's motion, ~2.5 km/s
- Coast a trans-Earth ellipse: apogee at the Moon, perigee grazing the atmosphere
- 3–5 day transit; arrive ~11 km/s (Apollo-class)
- PICA shield + parachutes, ocean / land recovery

A prograde shot would miss Earth and climb to a higher orbit instead — direction is the whole trick.

② CISLUNAR ORBITS (satellites & relays)

~15–20 × 100 kg ESPA-class smallsats per shot (≤7 t)

- Low lunar orbit: circularize at apolune (~0.3 km/s)
- EM-L1 halo, 58,000 km on the Earth-facing side
- EM-L2 / NRHO, 62,000 km past the far side (Gateway)
- Each sat adds its own small Δv to fine-tune

GEO transfer ~0.8 km/s; Earth LEO needs a ~3.1 km/s kick stage — it arrives at lunar-return speed, not orbital.

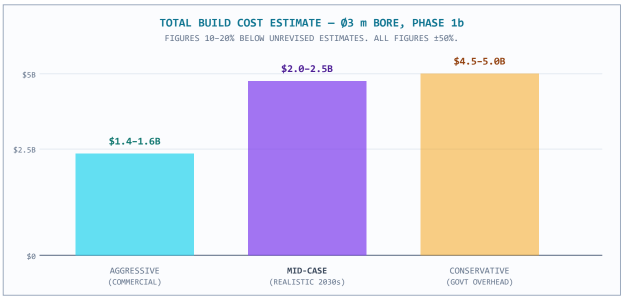
③ CISLUNAR DEPOTS (propellant, water, metal)

LOX/LH_2 · water · regolith shielding · metal ingots

- Aim the launch so apolune meets the depot's orbit
- Insertion / rendezvous burn ~150–300 m/s
- Berth or net-capture at an LLO or NRHO depot
- Transfer cargo; the pod flies back to reload

Delivered LOX / water ~$3.50/kg vs $3,000+/kg from Earth — this is the trade that funds the cislunar economy.

PANEL F, BUILD COST SUMMARY (REVISED ESTIMATE, Ø3 m, PHASE 1b)



COST CATEGORY BREAKDOWN (MID-CASE, Ø3 m)

FIXED (ALL BORE SIZES)

R&D + DESIGN + EARTH TEST $780M
SOLAR ARRAYS (8.7 MW, 43.5 t) MFG+DELIVERY $100M
CRYOGENIC PLANT (20 K, 15 t) $115M
CONSTRUCTION ROBOTS (20 t) $460M
POWER STORAGE (FLYWHEELS/CAPS, 30 t) $135M
ELECTRONICS + COMMS + CONTROL $215M
OPERATIONS (10-YR RUNWAY) $215M

VARIABLE (Ø3 m BORE)

REBCO TAPE (2,809 km @ $100/m) $281M
REBCO DELIVERY (6.7 t @ $2,000/kg) $13M
TRACK STRUCTURE, GUIDES, ALIGNMENT $110M
POD/BUCKET FLEET (10 UNITS) $100M

TOTAL MID-CASE ~$2.52B

COST / kg (10 yr × 72,000 t/yr = 720 Mt): ~$3.50/kg vs STARSHIP $200–2,000/kg

ALL FIGURES ±50%. REVISION FROM PRIOR ESTIMATE: 10–20% REDUCTION (ISRU CREDIT + COMMERCIAL MODEL + VOLUME REBCO PRICING)

SELENE DESIGN MAP (KER-SEL-TRK-002 REV E) · Ø3 m BORE · PHASE 1b PoC (2.9 km, 8.7 MW, 100 g, ESCAPE) · All physics derived June 2026, all hardware dimensions and cost preliminary. Real LRO/LOLA topography (NASA PDS, LDEM_875S_20M). © 2026 Perpetual Technologies. All rights reserved. Not for construction.